Ch 6.3: PCR

Lecture 20 - CMSE 381

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Wed, Oct 15, 2025

Announcements

Last time:

- PCA

This lecture:

- PCR

Announcements:

- Submit your questions by Thursday (10/16) 5 pm!
- Exam #2 on next week!
  - Bring 8.5x11 sheet of paper
  - Handwritten both sides
  - Anything you want on it, but must be your work
  - You will turn it in
  - Non-internet calculator
  - questions about project

12	F	9/26	Leave one out CV	5.1.1, 5.1.2	
13	M	9/29	k-fold CV	5.1.3	
14	W	10/1	More k-fold CV	5.1.4-5	
15	F	10/3	k-fold CV for classification	5.1.5	
16	M	10/6	Subset selection	6.1	
17	W	10/8	Shrinkage: Ridge	6.2.1	
18	F	10/10	Shrinkage: Lasso	6.2.2	HW #4 Due Sun 10/12
19	M	10/13	PCA	6.3	
20	W	10/15	PCR	6.3	
	F	10/17	***Review***		
	M	10/20	Fall Break		
	W	10/22	**Midterm #2**		
21	F	10/24	Polynomial & Step Functions	7.1-7.2	HW #5 Due Sun 10/26
22	M	10/27	Step Functions; Basis functions; Start Splines	7.2-7.4	

Section 1

Previously...

Shrinkage

Find β to minimize

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

subject to:

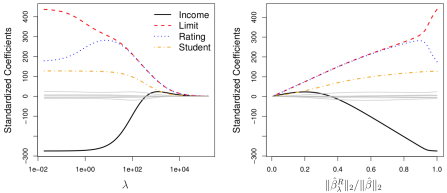
Least Squares:
No constraints

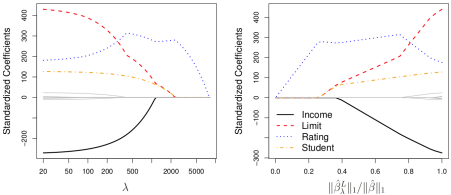
Ridge:

$$\sum_{j=1}^{p}\beta_j^2 \le s$$

The Lasso:

$$\sum_{j=1}^{p}|\beta_j| \le s$$





Also, find best choice of λ or s using CV

Linear transformation of predictors

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, Z_M$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

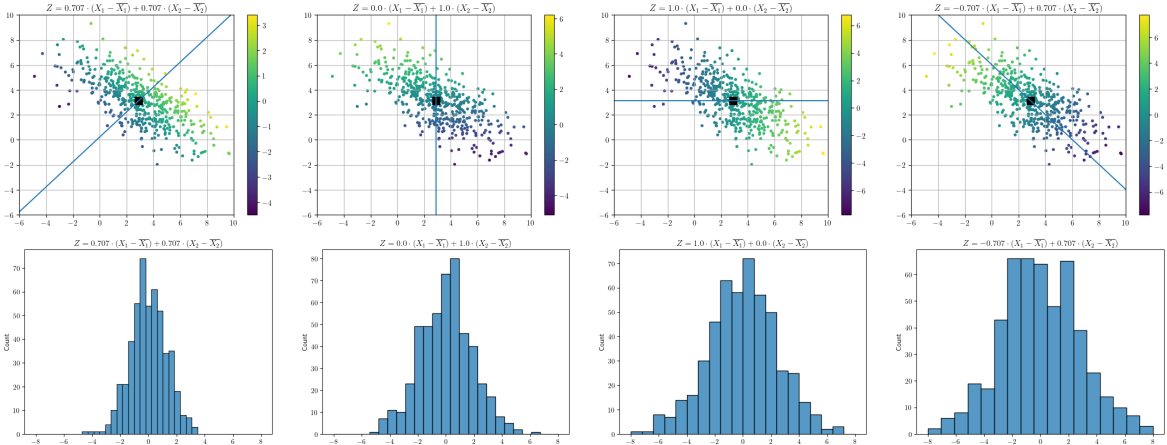
The goal:

- Find good φ's (PCA)
- Fit regression model on Z_i's using least squares (PLS)

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \varepsilon_i$$

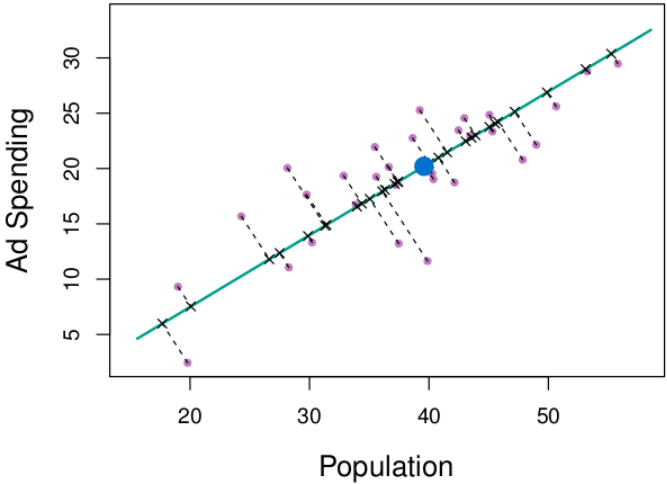
- Hope that lower dimensions means less overfitting

PCA - First PC



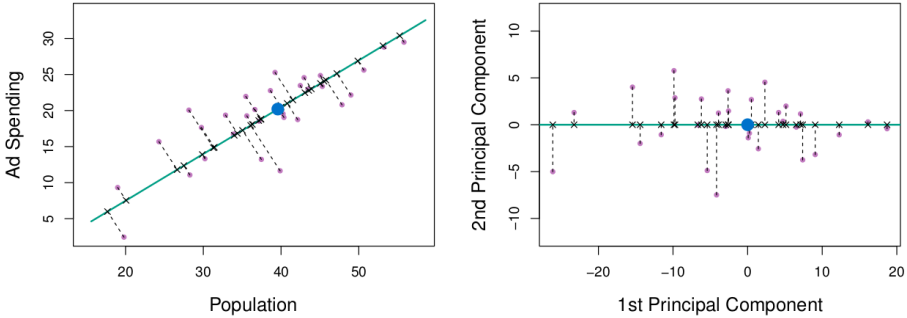
higher variance of projection on the further right pic

Projection onto first PC



$$Z_1 = 0.839 \cdot (\mathtt{pop} - \overline{\mathtt{pop}}) + 0.544 \cdot (\mathtt{ad} - \overline{\mathtt{ad}})$$

Drawing points in PC space



What will you learn from this lecture?

- Why do you want to use the PCs in regression models?
  - What assumptions do you have to make for it to be a good idea to use **principal component regression (PCR)**?
  - Or conversely, what is a typical bad scenario to use PCR?
- How do you implement PCR in Python?
- How do you interpret the model coefficients when using PCR?
- How do you choose the number of PC to use in PCR?
  - Given a figure of, e.g., cross-validation score as a function of the number of PCs, you should be able to choose appropriately and provide rationales in terms of bias-variance tradeoff.
  - You should be able to generate such figures in Python.
- What is the relationship/differences between PCR and feature selection and regularization methods that you learned in this part of the course?

Section 2

Principal Components Regression

So you've found your PCA coefficients

Now what?

- Do linear regression on $Z_1, \cdots, Z_M$
- Book calls this "principal components regression "
- Linear model is $y = \theta_0 + \theta_1 Z_1 + \cdots + \theta_M Z_M$

What are we assuming?

- The directions in which $X_1, \cdots, X_p$ have the most variation are the directions that are associated with Y
- If assumption holds, then fitting on Z_i better than fitting on X_i since fewer variables lessens chance of overfitting

Interpretation of PCR coefficients

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, Z_M$$

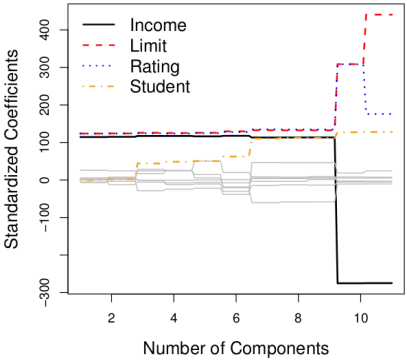
$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

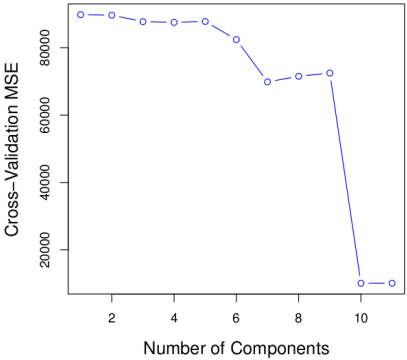
Learned model:

$$y = \theta_0 + \theta_1 Z_1 + \cdots + \theta_M Z_M$$

- Do example showing that coefficients of Z_i can't be used to say anything about X_i's
- $p = 3, M = 2$
- $\varphi = \begin{pmatrix} 1 & -1 \\ 3 & 1/2 \\ 0 & 1/2 \end{pmatrix}$
- Learned model $y = 2Z_1 - Z_2$
- $y = 2(X_1 + 3X_2 + 0X_3) - (-X_1 + \frac{1}{2}X_2 + \frac{1}{2}X_3)$
- $= 3X_1 + 2.5X_2 - 0.5X_3$
- Conclusion: can't directly use coefficients learned to interpret original inputs!
- Each PC is a linear combination of the originals, so can go backwards to figure out whaat the coefficients would have been, see left figure

Picking M





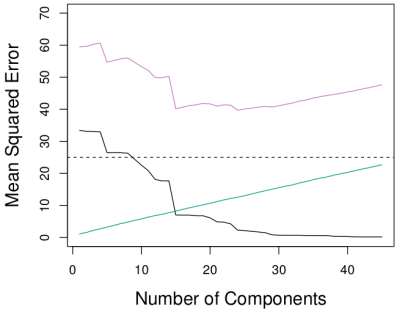
- Credit data set
- Major drop in MSE at 10 compoentns, so barely a savings with the original 11 variables

Do PCR with hitters data

Bias-Variance Trade-off

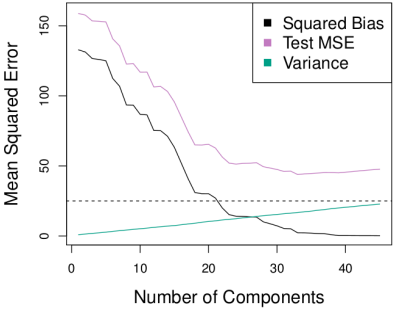
Example with simulated data: $n = 50$ observations of $p = 45$ predictors

Y is a function of **all predictors**



- Typical U shape for Test MSE
- Improve over least square (right most value x=45)

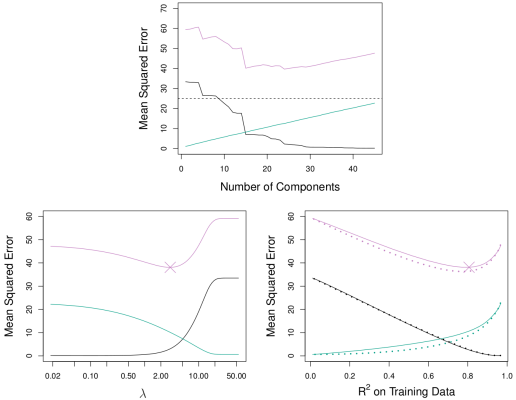
Y is a function of **2 predictors**



- Typical U shape for Test MSE
- (slight) improve over least square (right most value x=45)

Comparison to results on shrinkage

Y is a function of all predictors



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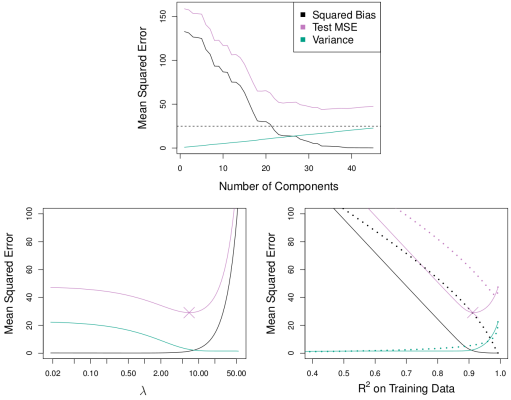


Shrinkage does better

Properties of PCR

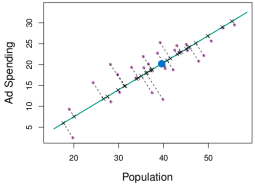
- Note better in previous example because lots of PCs needed
- It does better when info contained in the first few PCs
- Number of PCs to use can be picked with CV
- Standardization needed to make sure big data values don't skew the results
- Not a feature selection method because we don't get a subset of features, we get a collection of linear combos of features

Test your understanding: PollEv

TL;DR

PCR

- Unsupervised dimensionality reduction + linear regression
- Choose component Z_1 in the direction of most variance using only X_i's information
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained



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